Green Software Practitioner: Key Learnings

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1. Climate & Carbon Fundamentals

CO2eq

- **CO2eq:** Carbon equivalence metric, to normalise emissions to carbon.

Climate Goals & Institutions

- Paris Climate Act, IPCC (Intergovernmental Panel on Climate Change - UN Body for Climate impact policies), and other bodies:
  - Agree on climate goals.
  - Do annual follow-ups to assess trends.
  - Mobilise funds to developing nations.

Carbon Intensity

- **Carbon Intensity:** How carbon is emitted per kWh (gCO2eq/kWh).
- Carbon intensity varies with geographic availability.
- Renewable sources are susceptible to fluctuation (wind/solar).
- Hence, the grid will draw from coal/natural gas during a dip in clean sources.

2. Energy & Computing

Power Usage Effectiveness (PUE)

- **Power Usage Effectiveness (PUE):** Developed by Green Grid to quantify energy efficiency of data centres.
- Example:
  - Data Center consumes 10 kWh for computing.
  - PUE = 1.5.
  - Additional 5 kWh is consumed for cooling.
  - 15 kWh total.

Energy Proportionality

- **Energy Proportionality:** First proposed by engineers at Google.
- Relationship between power consumed by a computer and its utilisation.
- Utilisation, typically as a percentage, describes the usage of the computer's central resources.
- Relationship between power and utilisation is **NOT proportional**.
- The more you utilise your computer, the more energy efficient it becomes.

- Hence, for hardware efficiency, it's beneficial to run your systems on fewer servers.

Static Power Draw

- **Static power draw:** Electricity drawn when in an idle state.
- Unfortunately, servers are usually not designed with good power saving modes as their immediate availability is crucial for fast responsiveness to rapidly changing demands.

3. Renewable Energy & the Grid

Curtailment

- **Curtailment:** Disposal of excess energy when power source produces an excess.

Dispatchability

- **Dispatchability:** Ability to control the amount of energy produced.
- Fossil fuel sources are naturally more dispatchable:
  - Simply increase the amount of coal being burned.
- It's harder to control renewable sources such as the availability of wind.
- Renewable sources are actually cheaper and hence when demand surges, typically curtailed renewable sources are upped.
- Renewable energy is sometimes curtailed while fossil fuels are still consumed because:
  - Fossil fuel plants rarely scale down to 0.
  - They have a minimum functioning threshold.

4. Carbon-Aware Computing

Spatial Shifting

- **Spatial Shifting:** Moving computation to another physical location where carbon intensity is lower.
- Example:
  - Moving to different hemispheres depending on the season for more sunlight hours.

Temporal Shifting

- **Temporal shifting:** Find hours of the day to run software so as to optimise for carbon intensity (reduce it).
- Future carbon intensity can be predicted thanks to advancements in weather forecasting with ML.

Industry Examples

- Examples in industry of carbon awareness and demand shifting:
    - Google's carbon aware data centers.
    - Microsoft's carbon-aware Windows.

Demand Shaping

- **Demand Shaping:** Instead of shifting computation by location/temporally (Demand Shifting), Demand Shaping adjusts the demand of software running based on carbon availability.
- Higher carbon intensity → lower the demand.
- Vice versa.
- Example:
    - Eco Mode.

5. Embodied Carbon & Hardware Efficiency

Embedded / Embodied Carbon

- **Embedded/embodied Carbon:** Carbon pollution emitted during creation and disposal of a device.

Amortisation

- **Amortisation:** Accounting for the embodied carbon over the expected life span of a device.
- Example:
    - 4000 kg CO2eq to build a server.
    - Expected lifespan = 4 years.
    - Server emits 1000 kg CO2eq/year.

Hardware Efficiency

- Hardware Efficiency has 2 main approaches:
    1. **For end-user devices:** Extending the lifespan of the hardware.
    2. **For cloud computing:** Increasing the utilisation of the device.

Public vs Private Cloud

- All computation servers have peaks and troughs.
- When demand peaks, additional spare capacity is employed.
- Example:
    - An organisation averages at 20 servers but peaks at 100.
    - A private cloud of 100 will be needed.
- A shared public cloud enables **statistical multiplexing**:
    - Reducing the total number of servers to handle everyone's normal demands + peaks.

    - Peaks don't typically happen simultaneously.
- Capacity being shared across independent customers results in higher average utilisation.
- Shared infrastructure reduces overprovisioning.
- However, you need software architecture that can actually exploit that shared capacity.

6. Measurement & Software Carbon Intensity

Measurement

- **What you can't measure, you can't improve.**

GHG Protocol

- **GHG Protocol:** Most widely used and internationally recognised greenhouse gas accounting standard.
- 3 main scopes.

Software Carbon Intensity (SCI)

- While the GHG calculates total emissions, the **SCI is about calculating the rate of emissions**.
- Two buckets:
    - Operational emissions.
    - Embodied emissions.
- SCI score is reduced by investing time or resource into:
    - Energy efficiency.
    - Hardware efficiency.
    - Carbon awareness.

SCI Formula

- **SCI = ((E × I) + M) per R**

Where:

- **E** = Energy consumed by a software system.
- **I** = Location-based marginal carbon emissions.
- **M** = Embodied emissions of a software system.
- **R** = Functional unit (e.g. carbon per additional user, API-call, ML job, etc).
- **SCI = C per R (Carbon per R)**

7. SCI Calculation

Steps to Calculate SCI

1. **Decide what to include and exclude**
    - Software boundary and components.
2. **Choose functional unit**
    - Users.
    - Minutes.
    - ML jobs.
3. **Decide on how to measure emissions**
    - Counter-based measurement of energy usage.
    - Calculation using a model.
4. **Quantify SCI score.**

- Different SCI scores can be useful at once:
    - Different dimensions + view.

8. Carbon Reduction Methodologies

Abatement & Offsets

- **Carbon Reduction Methodologies:**
    1. Abatement / Carbon Elimination.
    2. Offsets.
- **Offsets:** Direct investments in emission-reduction projects through the purchase of carbon credits on the voluntary carbon market (VCM).
- The VCM is a decentralised market where private actors voluntarily buy and sell carbon credits that represent certified removals or reductions of GHGs from the atmosphere.
- **1 carbon credit → 1 tonne of CO2 absorbed or reduced.**

Offset 1: Compensating / Carbon Avoidance

- Essentially investing in other organisations' abatement projects.
- Examples:
    - Conservation.
    - Community projects.
    - Waste to energy.

Offset 2: Neutralisation / Carbon Removal

- By enhancing natural carbon sinks that remove CO2:
    - Forest reforestation.
    - Direct air capture (capturing CO2 from the air and storing permanently).

Durability

- Short, medium, long term carbon removal projects.

- **Durability** describes how long the carbon dioxide will be kept from the atmosphere.

9. Carbon Neutral vs Net Zero

Carbon Neutral

- **Carbon Neutral:**
  - All direct emissions (GHG scopes 1 and 2) are offset by either compensations or neutralisation efforts.
- While a significant step in encouraging orgs to measure their emissions, there are not enough offsets in the world to account for all emissions.
- Hence, abatement is needed for scaling and to achieve the 1.5C target by Paris climate agreement.

Net Zero

- **Net Zero:**
  - Reduce emissions according to latest climate science.
  - Balance remaining emissions through carbon removals.
- In alignment with 1.5C pathway.
- All businesses must do this to achieve net-zero emissions by 2050.
- Net Zero focuses on abatement rather than offsets.
- Targets are set by the Science Based Targets Initiative (SBTi).
- Target is **90% eliminations of carbon emissions by 2050**.

10. Renewable Energy & 24/7 Matching

100% Renewable

- **100% Renewable → Powered vs Matched.**

Renewable Energy Certificates (RECs)

- Renewable plant sells its clean energy as well as Renewable Energy Certificates.
- **1 REC = 1 kWh of energy.**
- To be 100% matched by renewable energy while being on the grid:
  - Only solution is to buy enough RECs to cover the amount of electricity you consume.

Power Purchase Agreements (PPAs)

- **Power Purchase Agreement:** Another way to purchase RECs.
- Long-term contracts with a renewable plant.
- Also a financing source for said plant.
- Purchasing PPAs drives the creation of new plants.

- **Additionality.**

24/7 Hourly Matching

- **24/7 Hourly Matching:** 100% matched by renewable energy hourly 24/7.
- Google and Microsoft have committed to this by 2030.

Carbon-Free Energy (CFE)

- **Carbon-free energy (CFE):** How successful we are at 24/7 matching.

11. Final Takeaway

Software needs to be **carbon-aware** to achieve CFE goals. By improving energy and hardware efficiency, increasing utilisation, measuring Software Carbon Intensity, and shifting computation spatially and temporally based on carbon intensity, software can reduce its emissions. As renewable energy becomes more available, software can help match demand with clean energy and move toward **24/7 carbon-free energy**.